Dataset Selection and Suitability Analysis

Governed Multi-Agent AI for Autonomous E-Commerce and Accounting Operations

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Abstract

This report evaluates the local Olist Brazilian E-Commerce dataset and Online Retail II for a governed multi-agent e-commerce and accounting prototype. All nine Olist CSV tables and the two worksheets in the local Online Retail II workbook were inspected without modifying source files. Olist comprises 9 tables, including 99,441 orders, 112,650 order-item rows, 103,886 payment rows, and 32,951 products. Online Retail II contains 1,067,371 line records in two annual worksheets and the expected eight transaction fields. Olist supports broader linked order and marketplace workflows; Online Retail II provides a much larger product-line transaction history. Neither dataset provides complete inventory movements, a general ledger, explicit governance policies, or verified fraud labels. The datasets are therefore suitable for transaction replay, rule-based prototype checks, and auditable derived summaries, subject to careful handling of missingness, duplicate rows, and return/cancellation semantics. They must be integrated through independent source adapters; actual records must not be joined across datasets.

1 Introduction and research context

The project studies a prototype with orchestration, e-commerce, inventory, accounting, risk, governance, and audit agents. Public transaction datasets can supply evidence for replayable workflows, but they do not by themselves instantiate agents or business controls. This assessment separates observable source fields from derived analyses, external requirements, and prototype behavior.

2 Objectives and method

The objective is to select and characterize locally available data for the research prototype. The inspection script recursively scanned `datasets/` for CSV, Excel, and JSON files, read CSVs as UTF-8 with BOM support and detected delimiters, and processed the Excel workbook as OOXML. It calculated dimensions, field names, missingness, distinct non-empty values, numeric ranges, exact duplicate rows, and selected cross-table key containment checks. All reported counts are based on local files and are reproducible from the included script and machine-readable artifacts. Numeric-looking codes are not automatically treated as measures. The comparison uses qualitative ratings: High means direct, broad field support for the stated prototype task; Medium means useful but partial fields or substantial derived logic; Low means a material core capability is absent; Unknown means evidence or terms were not verified. For licensing, High indicates a verified source license that permits reuse with stated conditions; Medium indicates verified terms with meaningful use conditions; Unknown means the license was not verified. These are task-relative judgments, not universal dataset rankings.

3 Local availability and inspection environment

The local workspace contains nine Olist CSV files (total 126,186,995 bytes; individual byte counts are in Table 1) and a 45,622,278-byte workbook at `datasets/online_retail_ii/online_retail_II.xlsx`. The workbook has sheets `Year 2009-2010` (525,461 data rows) and `Year 2010-2011` (541,910 data rows). Python 3 is available. Pandas and openpyxl were absent, so the included inspector uses the standard library. MiKTeX `pdflatex` and BibTeX are installed. The Excel date values are stored as spreadsheet serials; workbook dates are interpreted as timestamps for the date-range statement, while the low-level quality CSV preserves the underlying numeric serial range.

Table 1: Locally observed Olist file inventory. Dimensions exclude header rows.

File	Rows	Columns	Bytes
customers	99,441	5	9,033,957
geolocation	1,000,163	5	61,273,883
orders	99,441	8	17,654,914
order items	112,650	7	15,438,671
order payments	103,886	5	5,777,138
order reviews	99,250	7	14,451,670
products	32,951	9	2,379,446
sellers	3,095	4	174,703
category translation	71	2	2,613

4 Olist dataset: structure and empirical findings

The observed schema consists of customer location records; geolocation; order status/timestamps; item-level product, seller, price, and freight; payment method/installments/value; review score/comments/timestamps; product dimensions/category; seller location; and a category translation table. The files provide a relational view, rather than one flat fact table. The `order_id` and `customer_id` columns each have 99,441 distinct non-empty values in their respective 99,441-row tables; `product_id` has 32,951 distinct values in 32,951 product rows and `seller_id` has 3,095 distinct values in 3,095 seller rows. These candidate keys are unique by observed cardinality. The customer table has 96,096 distinct `customer_unique_id` values, showing that this field identifies repeat customer groupings rather than a one-row-per-order key. Olist has 99,441 order identifiers and 99,441 customer records. The candidate customer and order keys have no duplicate key values, as confirmed by the key-check artifact. The order-item table has 112,650 rows, while its 98,666 distinct order IDs are all present in orders. The payment table has 99,440 distinct order IDs, all present in orders. The review table has 98,690 distinct order IDs; 17 distinct keys do not occur among the order IDs. These checks are calculated on observed non-empty identifiers and do not mean that every order has every kind of child record. Candidate key checks further show that the composite (`order_id`, `order_item_id`) is unique across 112,650 item rows and (`order_id`, `payment_sequential`) is unique across 103,886 payment rows; the single `order_id` field is not unique in those child tables. These duplicate-excess and missing-key checks are provided in `primary_key_checks.csv`. Reviews have no separate review identifier in the observed schema, so no review primary key is asserted. Order items expose 32,951 distinct product IDs and 3,095 distinct seller IDs, all contained in the products and sellers keys respectively.

Olist item prices range from 0.85 to 6,735.00 and freight values range from 0.00 to 409.68 in the source currency (Brazilian real is implied by the Brazilian marketplace context; the CSV itself has no explicit currency-code field). Payment values range from 0.00 to 13,664.08.

These are observed extrema, not cleaned or adjudicated accounting values. Observed timestamp ranges include order purchase from 2016-09-04 21:15:19 to 2018-10-17 17:30:18; carrier handoff from 2016-10-08 10:34:01 to 2018-09-11 19:48:28; customer delivery from 2016-10-11 13:46:32 to 2018-10-17 13:22:46; estimated delivery from 2016-09-30 to 2018-11-12; review creation from 2016-10-02 to 2018-08-31; review answer from 2016-10-07 to 2018-10-29; and item shipping limits from 2016-09-19 to 2020-04-09. These are observed field ranges, not necessarily order lifecycle bounds. The date columns include purchase, approval, carrier handoff, delivery, estimated delivery, review creation/answer, and shipping limit. Missing order timestamps include 160 approval timestamps (0.161%), 1,783 carrier timestamps (1.793%), and 2,965 customer delivery timestamps (2.982%). Status and timing must therefore be interpreted with missingness and order status in view.

The geolocation file has 1,000,163 rows and 261,831 exact duplicate rows. Exact duplicate rows are observed as full-row repetition; repeated postal prefixes by themselves are not necessarily errors. Product category and descriptive dimensions are incomplete: 610 product category/name-length/description-length/photo-count entries are missing (1.851% of 32,951); two products lack each of several physical dimensions (0.006%). Review text is especially sparse: 87,680 titles (88.343%) and 58,299 messages (58.740%) are empty. Ten review scores and nine review order IDs are missing. Percentages use the row count of the corresponding table as denominator. Full per-column values are supplied in `data_quality_summary.csv`.

5 Online Retail II: structure and empirical findings

The workbook has eight fields: `Invoice`, `StockCode`, `Description`, `Quantity`, `InvoiceDate`, `Price`, `Customer ID`, and `Country`. It contains 1,067,371 data rows across two sheets, with 12,133 exact duplicate line rows detected in the raw cell values (6,865 and 5,268 by worksheet). Negative quantities occur in 22,950 rows (12,326 and 10,624 by sheet), with no zero quantities. There are 19,494 rows whose invoice identifier has the documented cancellation prefix (10,206 and 9,288 by sheet), representing 4,592 and 3,836 distinct cancellation invoice identifiers within each worksheet. Zero prices occur in 6,202 rows (3,687 and 2,515), and five rows have negative prices (three and two). These counts identify records for contextual review; they are not fraud or error determinations. The first sheet has 2,928 missing descriptions (0.557%) and 107,927 missing customer IDs (20.540%). The second has 1,454 missing descriptions (0.268%) and 135,080 missing customer IDs (24.927%). Invoice, stock code, quantity, invoice date, unit price, and country are populated throughout the inspected rows. A customer identifier therefore cannot be assumed for every transaction.

The source's date values span 1 December 2009 to 9 December 2011, consistent with UCI's dataset description. Each worksheet has 8 columns. The first contains 28,816 distinct invoice cell values, 4,631 stock-code values, 40 countries, and 4,383 non-empty customer IDs. The second contains 25,900 invoice values, 4,070 stock codes, 38 countries, and 4,372 non-empty customer IDs. Because invoice identifiers are present in both sheets and the same invoice value might recur across partitions, totals across sheets require global recomputation rather than adding the sheet-level distinct counts. Quantity ranges are -9,600 to 19,152 (first sheet) and -80,995 to 80,995 (second). Price ranges include negative and unusually large values: -53,594.36 to 25,111.09 and -11,062.06 to 38,970.00. These values merit review and are not interpreted as errors without source business context. Invoice IDs beginning with `C` are a documented cancellation marker; that is a transaction status convention, not evidence of fraud. Negative quantities may represent returns or corrections; zero/negative or very large prices may reflect adjustments, special lines, or data-quality issues. The inspection artifacts report raw-value ranges and do not silently filter them.

6 Data quality and schema interpretation

For CSVs, blank fields were counted as missing. For Excel, blank cells were counted within each worksheet. Exact duplicate detection compares complete decoded field values (or workbook cell values) and counts repeated rows after their first occurrence. The high geolocation duplicate count is salient and should be assessed in a future pipeline using key semantics and source documentation, not automatically removed. Identifiers must be handled as strings even where they contain digits; quantity and price are numeric measures, while invoice dates require proper spreadsheet date decoding. The included quality file provides missing count and percent, distinct count, and numeric extrema by field. Date values stored as Excel serials are not misreported as calendar timestamps in that raw extrema table.

7 Dataset comparison

Table 2 applies the defined qualitative scale. Reproducibility is High for both in this workspace because the original files are present and all analyses use local inputs. Licensing is evaluated separately from local accessibility. Ratings reflect the intended multi-agent prototype, not dataset merit in general.

8 Agent-to-dataset suitability

Capabilities are marked as direct (source columns), derived (computed from columns), external (requires data not present), or prototype (must be implemented as agent/rule behavior). The table describes an achievable prototype scope, not native automation embedded in either dataset.

Table 3: Agent suitability and implementation boundary.

Agent	Olist	Online Retail II	Available evidence	Missing information	Proposed implementation
Orchestrator	Prototype / Medium	Prototype / Low	Transaction rows can be replayed through task workflows	No agent events or coordination trace	Simulate agent messages, state transitions, and policy gates.
E-Commerce	Direct / High	Direct / High	Olist orders, status, customers, and payments; Retail II invoices and product-line fields	Retail II lacks payment method and order lifecycle status	Separate order and invoice scenarios in adapters.
Inventory	Derived / Medium	Derived / Medium	Olist item quantities and product IDs; Retail II quantities and stock codes	External: stock on hand, receipts, transfers, full returns	Derive sales/return signals; simulate inventory balances with declared assumptions.
Accounting	Derived / Medium	Derived / Medium	Olist item prices, freight, and payment values; Retail II quantity and unit price	External: tax, settlement, currency metadata, chart of accounts, ledger	Reconcile transaction evidence; label all totals as analytical, not ledger postings.

Agent	Olist	Online Retail II	Available evidence	Missing information	Proposed implementation
Risk and Anomaly Detection	Derived / Medium	Derived / Medium	Amount, status, timing, quantity, identifier consistency	External: adjudicated fraud labels and contextual risk features	Apply transparent anomaly rules; report flags for review, not fraud determinations.
Governance	Prototype / Medium	Prototype / Medium	Fields support explicit threshold and completeness checks	External: organizational policies, approvals, retention/access rules	Version rules as prototype configuration and log outcomes.
Audit	Derived / Medium	Derived / Medium	Stable source IDs and repeatable table-level validation	No native chain-of-custody or agent action log	Preserve source keys, validation outputs, and run metadata.

9 Independent adapters and canonical schema

The datasets must remain logically independent. The primary design is one adapter per source, producing a shared event interface without joining records. A minimal canonical event may contain `source_dataset`, `source_record_key`, `event_kind`, `event_date`, `item_key`, `quantity`, `unit_amount`, and `currency_code`. `source_dataset` and source keys are mandatory lineage. Olist can map `order_id` to an order event, `order_item_id` to a line key, `product_id` to an item key, `price` and `freight_value` to separate amounts, and payment values to payment events. The currency code is not present and must be configured or left unavailable. Retail II maps invoice, stock code, quantity, invoice date, and unit price; invoice cancellation status can be derived from the documented `C` prefix. Customer IDs are optional. A canonical customer key remains source-scoped. Missing values are represented as null with an explicit `Not available in source dataset` explanation, never fabricated.

A second, optional layer can normalize event/line concepts and aggregation units for comparative experiments. It is schema normalization only. The numeric order/invoice IDs, product codes, customer identifiers, countries, and date semantics are source-specific and must never be treated as common business identities. Any mapping beyond the direct field correspondences above requires an explicit versioned assumption.

10 Accounting and inventory boundaries

Olist contains item prices, freight amounts, payment values, payment methods, and installment counts. This supports candidate transaction-level summaries and consistency checks, but rows are not a complete double-entry general ledger: there is no verified account mapping, tax/fee structure, settlement statement, or general-ledger posting history. Retail II offers quantity and unit price for invoice lines but does not provide payment/settlement data or explicit currency code; UCI documents its unit prices in sterling. Neither dataset tracks stock-on-hand over time, receipts, transfers, reservations, or a complete return-to-stock process. Inventory balances must therefore be simulated or supplied from an external inventory source.

Table 2: Qualitative comparison against the prototype objectives.

Criterion	Olist	Online Retail II	Evidence and qualification
Research/workflow relevance	High	High	Both contain real transaction records; Olist spans more linked workflow entities, while Retail II offers a longer line-level series.
E-commerce coverage	High	Medium	Olist includes orders, status, customers, items, payments, sellers, and reviews; Retail II contains invoices and product lines.
Inventory evidence	Medium	Low	Both support item/product quantity analysis; neither contains historical on-hand stock or full movement ledgers.
Accounting evidence	Medium	Medium	Item amounts, freight (Olist), unit prices (Retail II), and payment records (Olist) support reconciliations; neither is a general ledger.
Risk/anomaly analysis	Medium	Medium	Both permit rule-based outlier and consistency checks; neither has verified fraud labels.
Governance/audit support	Medium	Medium	Reproducible rules and traceable rows can be built; policies and agent decisions must come from prototype configuration.
Data quality/completeness	Medium	Medium	Both have meaningful missingness and unusual values; Olist geolocation has many exact duplicate rows and Retail II has missing customer IDs and repeated lines.
Integration effort	Medium	Medium	Requires two adapters and source-specific semantics; there is no safe record-level cross-source join.
Local reproducibility	High	High	Nine Olist CSVs and the workbook are present and inventoried.
License/redistribution	Unknown	High	UCI record states CC BY 4.0 and requires attribution. Olist primary Kaggle listing was verified, but its license terms were not independently confirmed from the accessible data card.

11 Risk, governance, audit, and coordination limits

Unusual amounts, negative quantities, cancellations, missing identifiers, and referential anomalies can trigger review rules. A flagged row is not a confirmed fraud case. Neither inspected schema provides a verified fraud outcome label. Governance policies (thresholds, approval authority, segregation of duties, privacy retention) are not source data and must be explicitly defined and versioned by the prototype. Auditability can be implemented through repeatable source-key checks, immutable input manifests, run identifiers, and agent decision logs; these capabilities are not automatically present in the files. Orchestration behavior is entirely prototype logic. Historical data can drive deterministic replays, but it cannot evidence autonomous coordination unless the prototype emits and evaluates its own traces.

12 Provenance, licensing, and usage

The Olist listing on Kaggle identifies the dataset title/provider, describes approximately 100,000 orders, and lists nine CSV files [2]. Its current license terms were not independently confirmed from the accessible primary data card during this inspection; verify those terms before redistributing data or derived artifacts that reproduce source content. Secondary source claims about a specific Olist license are not treated as a verified permission here. UCI identifies Online Retail II as a dataset by Daqing Chen, describes its two-year transaction scope, provides the workbook, DOI, and states CC BY 4.0 [1]. Attribution is required under that license. This report does not provide legal advice; users should retain dataset attribution and check current terms before redistribution.

13 Recommended prototype configuration

Use Olist as the primary end-to-end workflow fixture because its local relational tables jointly cover order status, customers, order lines, payments, products, sellers, and reviews. Use Online Retail II as an independent transaction-line replay fixture for quantity/price patterns, cancellations, and return-like observations. Operate each through a source-specific adapter and shared, source-scoped event interface. Implement inventory balances, accounting ledgers, governance policy checks, risk thresholds, audit trails, and orchestration as explicit prototype modules; label simulated outputs. Preserve raw inputs and document each derived metric and exclusion.

14 Threats to validity and limitations

The inspected files may not represent the provider's latest release. Field typing is a heuristic, and source semantics can exceed what column names establish. Exact duplicate counts depend on decoded cell representation; duplicates are not automatically erroneous. The distinct-key joins test membership, not row-level referential integrity in every direction, cardinality constraints, temporal validity, or business completeness. Excel serial timestamps require workbook date-format interpretation. License verification is stronger for the UCI file than for Olist. Qualitative ratings depend on prototype objectives and should be revisited if the required task changes. No cross-source business identity is established.

15 Conclusion

Both local datasets are usable for governed multi-agent workflow prototyping when their evidence boundaries are respected. Olist provides the broader relational workflow context, while Online Retail II contributes a larger two-year line-level invoice history. A dual-adapter configuration preserves each source's semantics and supports controlled comparative replays without asserting false equivalence. Neither source supplies the stock history, complete accounting records, explicit governance rules, confirmed fraud labels, or agent coordination needed to claim production autonomy; these must be externally provided or explicitly simulated and audited.

References

[1] Daqing Chen. Online retail ii. `https://archive.ics.uci.edu/dataset/502/online+retail+ii`, 2012. UCI repository record; CC BY 4.0 stated on the official record.

[2] Olist. Brazilian e-commerce public dataset by olist. `https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce`. Kaggle listing; release and license metadata not independently verified.